

Customer Lifetime Value (LTV) Prediction Using Machine Learning

Online Retail Dataset | XGBoost Regression | Customer Segmentation

Introduction

Customer Lifetime Value (LTV) prediction helps businesses estimate the total revenue a customer may generate over time. It is widely used in customer analytics, targeted marketing, and customer retention strategies. In this project, machine learning techniques were used to analyze customer purchasing behavior and predict future customer value using the Online Retail dataset. Customers were also segmented into different value categories based on predicted LTV.

Abstract

This project focuses on predicting Customer Lifetime Value (LTV) using Machine Learning and the Online Retail Kaggle dataset. The dataset was preprocessed by removing missing values, cancelled invoices, and invalid transactions. Important customer behavior features such as Recency, Frequency, Monetary Value, and Average Order Value (AOV) were created using feature engineering techniques.

The XGBoost Regression algorithm was used to train the prediction model. The model performance was evaluated using MAE, RMSE, and R^2 Score. Customers were segmented into Premium, High Value, Medium Value, and Low Value categories based on predicted LTV values.

The final project generated customer segmentation visualizations, a trained machine learning model, and a CSV file containing predicted customer lifetime values. The project demonstrates the use of predictive analytics for targeted marketing and customer retention strategies.

Tools Used

Tool	Purpose / Role in Project
Python	Core programming language for data processing and model building
Pandas	Data manipulation, cleaning, and feature engineering
NumPy	Numerical computations and array operations
Matplotlib	Static data visualizations and plotting
Seaborn	Statistical data visualization with enhanced aesthetics
Scikit-learn	Data splitting, feature scaling with StandardScaler
XGBoost	Gradient boosting regression model for LTV prediction
Joblib	Saving and loading the trained machine learning model
Jupyter Notebook	Interactive development environment for experimentation

Steps Involved in Building the Project

Step 1: Dataset Collection

The Online Retail dataset was downloaded from Kaggle. The dataset contains customer transaction details such as invoice number, purchase date, quantity, unit price, and customer ID.

Step 2: Data Cleaning

The dataset was cleaned by removing missing values, cancelled invoices, negative quantities, and invalid prices to improve data quality.

Step 3: Feature Engineering

Important customer behavior features such as Recency, Frequency, Monetary Value, and Average Order Value were created using transaction history.

Step 4: Creating Target Variable

Customer Lifetime Value (LTV) was calculated and used as the target variable for machine learning prediction.

Step 5: Data Splitting and Scaling

The dataset was divided into training and testing datasets. Feature scaling was performed using StandardScaler.

Step 6: Model Training

The XGBoost Regression model was trained using customer behavioral features to predict customer lifetime value.

Step 7: Model Evaluation

The performance of the model was evaluated using MAE, RMSE, and R^2 Score to measure prediction accuracy.

Step 8: Customer Segmentation

Customers were segmented into Premium, High Value, Medium Value, and Low Value categories based on predicted LTV.

Step 9: Visualization and Export

Graphs and visualizations were created to represent customer insights. Final predicted LTV results were exported into a CSV file.

Conclusion

The Customer Lifetime Value Prediction project successfully demonstrated how machine learning can be used to analyze customer behavior and predict future customer value. The XGBoost Regression model provided accurate LTV predictions and helped classify customers into different value segments. This project shows how predictive analytics can improve marketing strategies, customer retention, and business profitability.